TOBB ETÜ

University of Economics & Technology

Proposal

Final Stack

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1. Project Title :

AI-Enhanced Data Preprocessing and Ingestion Layer

2.Problem Definition and Objectives:

Problem Definition :

In Neonatal Intensive Care Units (NICUs); medical devices, sensors, and Hospital Information Systems (HIS) often operate independently and use incompatible data formats, which prevents clinicians from obtaining a unified, real-time view of newborns' medical conditions. Moreover, raw data from sensors are frequently noisy, incomplete, or non-standardized; which reduces the accuracy and reliability of AI-based diagnostic and monitoring systems.

Clinicians also lack an integrated platform that consolidates historical patient records, AI-generated diagnostic results, and updated clinical guidelines. As a result, clinical decision-making becomes slower, less consistent, and more prone to human error.

To address these issues, there is a clear need for a secure, interoperable, and real-time digital decision support infrastructure that can unify all NICU data sources—IoMT devices, HIS platforms, and AI diagnostic modules—under standardized protocols such as HL7, DICOM, and FHIR.

Objectives:

1. **Develop an integrated digital decision support system** for Neonatal Intensive Care Units (NICUs) that unifies data from IoMT devices, Hospital Information Systems (HIS), and AI diagnostic modules.

2. **Implement standardized data communication protocols** (HL7, DICOM, FHIR) to ensure secure and interoperable information exchange across all hospital systems.

3. **Design a real-time data preprocessing pipeline** to clean, normalize, and structure physiological and imaging data for accurate AI analysis and clinical interpretation.

4. **Provide an intuitive web-based dashboard** that visualizes real-time patient data, AI predictions, and historical trends to assist clinicians in diagnosis and treatment planning.

5. **Establish robust security and access control mechanisms** to guarantee compliance with HIPAA and GDPR while protecting sensitive neonatal patient data.

6. **Enable knowledge sharing and case-based learning** by integrating AI-generated reports, historical patient records, and clinical guidelines into a searchable knowledge repository.

3.Scope:

- **Data Acquisition and Integration:**
  Real-time collection of physiological, imaging, and signal data from IoMT-enabled NICU devices (e.g., incubators, ventilators, pulse oximeters) and synchronization with Hospital Information Systems (HIS).

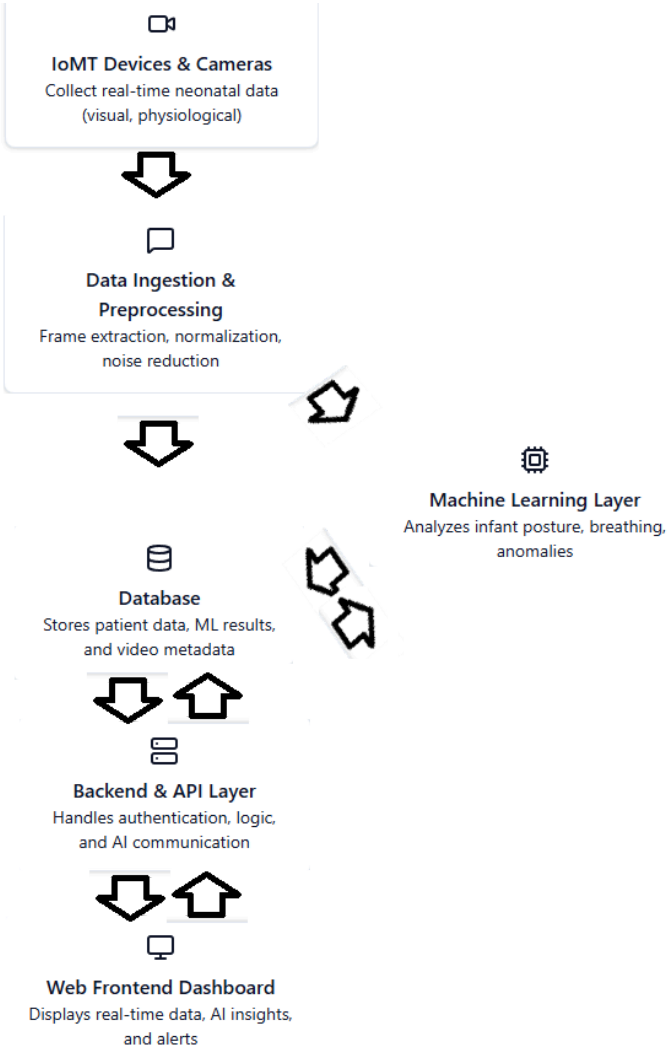
- **Data Preprocessing and Standardization:**
  Implementation of an AI-enhanced preprocessing pipeline for cleaning, normalizing, and transforming raw data into standardized formats (HL7, DICOM, FHIR).

- **AI Diagnostic and Decision Support Layer:**
  Integration of machine learning modules for automated analysis of neonatal health indicators, anomaly detection, and predictive alerts.

- **Web-Based Visualization and Interaction:**
  Development of a responsive dashboard allowing clinicians, nurses, and administrators to visualize patient data, AI predictions, and historical records in real time.

- **Knowledge Management System:**
  Storage and retrieval of case histories, AI-generated reports, and neonatal clinical guidelines, enabling pattern search and evidence-based recommendations.

- **Security and Access Control:**
  Role-based user authentication, data encryption, and audit logging to ensure compliance with HIPAA/GDPR and institutional policies.



4. User Profile

- AI Researchers
- Data Engineers
- NICU Clininicians
- Clinic IT/Adminstration

5. Anticipated Challenges & Constraints

- **Technical Hurdles:**
    - Real time data processing
    - Working with parallel data streams
- **Time/Resource Limits:**
    - Changes on data sources and AI Models over time
- **Ethical or regulatory considerations:**

    - Working with patient health data has legal constraints like GDPR

6. Economic, social, environmental, legal, health, security impacts

In Neonatal Intensive Care Units (NICUs), the disparate nature of medical device data, sensor readings, and Hospital Information Systems (HIS) presents a significant challenge. This fragmentation hinders clinicians from achieving a comprehensive, real-time understanding of a newborn's medical status. Consequently, the efficacy of AI-driven diagnostic and monitoring systems is diminished, and the speed and consistency of clinical decision-making are adversely affected. Therefore, there is a critical need for a secure, integrated, and real-time digital support infrastructure capable of unifying all NICU data sources under standardized communication protocols.

7. Data Sources

- IoMT device data (respiratory, cardiac).
- Ultrasound images (DICOM).
- Pulse oximetry signals.